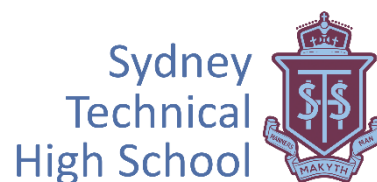


Student Name _____

Teacher's Name: _____



Extension 2 Mathematics

TRIAL HSC

August 2020

General

Instructions

- Reading time – 10 minutes
- Working time – 180 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- In questions 11-16, show relevant mathematical reasoning and/or calculations

Total marks:
100

Section I – 10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks

- Attempt questions 11-16
- Allow about 2 hours and 45 minutes for this section

Section I

10 Marks

Attempt Questions 1-10

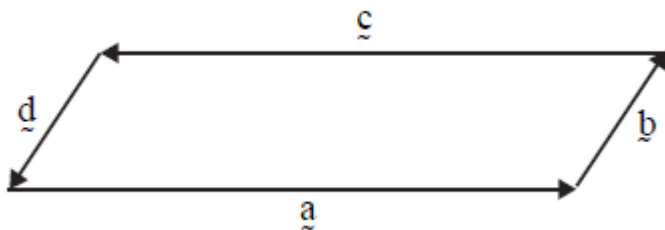
Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1-10.

1. If $z = 2e^{-\frac{\pi}{3}i}$, then which of the following is purely real?
- (A) $\bar{z}$
(B) z^3
(C) z^i
(D) $3z$
2. A particle's acceleration $\ddot{x} \text{ ms}^{-2}$ is defined by $\ddot{x} = -3x$, where x is the displacement in metres. How long, in seconds, does it take to travel between the endpoints of the motion?

- (A) $\frac{\pi\sqrt{3}}{3}$
(B) $9\pi\sqrt{3}$
(C) $3\pi\sqrt{3}$
(D) $\pi\sqrt{3}$

3. In the parallelogram, $|\underline{a}| = 2|\underline{b}|$



Which one of the following statements is true?

- (A) $\underline{a} = 2\underline{b}$
(B) $\underline{a} + \underline{b} = \underline{c} + \underline{d}$
(C) $\underline{a} + \underline{c} = 0$
(D) $\underline{a} - \underline{b} = \underline{c} - \underline{d}$

4. If $z = 3 - 4i$, then $\frac{1}{1-z}$ is equal to:

(A) $\frac{-1 - 2i}{10}$

(B) $\frac{-1 + 2i}{10}$

(C) $\frac{-1 - i}{6}$

(D) $\frac{-1 + i}{6}$

5. Assume that a and b are negative real numbers with $a > b$. Which of the following might be false?

(A) $\frac{1}{a-b} < 0$

(B) $\frac{a}{b} - \frac{b}{a} < 0$

(C) $a + b > 2b$

(D) $2a > 3b$

6. If the vectors $\vec{a} = m\vec{i} + 4\vec{j} + 3\vec{k}$ and $\vec{b} = m\vec{i} + m\vec{j} - 4\vec{k}$ are perpendicular, then:

(A) $m = -6$ or $m = 2$

(B) $m = -2$ or $m = 6$

(C) $m = -2$ or $m = 0$

(D) $m = -1$ or $m = 1$

7. Suppose that both x and y are odd. Which of the following statements is true?

(A) $x + y$ is odd

(B) $x - y$ is even

(C) $3x + 5y$ is odd

(D) xy is even

8. Using a suitable substitution, $\int_1^{e^3} \frac{(\ln(x))^3}{x} dx$ may be expressed completely in terms of u as:

(A) $\int_0^3 \frac{u^3}{e^u} du$

(B) $\int_0^{e^3} u^3 du$

(C) $\int_0^3 u^3 du$

(D) $\int_1^{\ln(3)} u^3 du$

9. Let the complex number z satisfy the equation $|z + 4i| = 3$. What are the greatest and least values of $|z + 3|$?

(A) 8 and 2

(B) 5 and 2

(C) 8 and 3

(D) 8 and 5

10. P, Q and R are three collinear points with position vectors $\underline{p}, \underline{q}$ and $\underline{r}$ respectively. Q lies between P and R . If $2|\overrightarrow{QR}| = |\overrightarrow{PQ}|$, then $\underline{r}$ is equal to:

(A) $\frac{3}{2}\underline{q} - \frac{1}{2}\underline{p}$

(B) $\frac{3}{2}\underline{p} - \frac{1}{2}\underline{q}$

(C) $\frac{1}{2}\underline{p} - \frac{3}{2}\underline{q}$

(D) $\frac{1}{2}\underline{q} - \frac{3}{2}\underline{q}$

Section II

Total marks – 90

Attempt Question 11-16

Allow about 2 hours and 45 minutes for this section

Begin each question on a NEW page

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Begin a NEW page.

- a) What is the unit vector that has the same direction as $\vec{v} = \begin{pmatrix} 6 \\ 3 \\ -1 \end{pmatrix}$ 2
- b) If $z = 6e^{i\frac{\pi}{3}}$
- i. Rewrite z in modulus argument form 1
 - ii. Simplify $(z)^4$ 2
 - iii. Find $\arg(iz)$ 2
- c) Integrate:
- i. $\int \frac{x+1}{x^2+2x+5} dx$ 2
 - ii. $\int \frac{1}{x^2+2x+5} dx$ 2
- d) Popularised on the internet, different types of dogs are given “names”. Three of these names are given as the following propositions.
- p : it is not a doggo
 q : it is a floofer
 r : it is a woofier
- i. Write down in words, $p \Rightarrow \text{not } r$ 2
 - ii. Write down the converse of $p \Rightarrow (\text{not } q \text{ and } r)$ 1
 - iii. Write down the contrapositive of $p \Rightarrow \text{not } q$ 1

End of Question 11

Question 12 (15 marks) Begin a NEW page.

- a) Consider the complex numbers $z_1 = 1 + i$ and $z_2 = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$
- i. Write z_1 in the form $r(\cos\theta + i\sin\theta)$ **2**
 - ii. Find the modulus of $z_1 z_2$ **1**
 - iii. By considering the expansion of $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$, or otherwise, write $z_1 z_2$ in the form $z = a + bi$ where $a, b \in \mathbb{R}$ **3**
 - iv. Hence find the value of $\tan\frac{5\pi}{12}$ in the form $c + d\sqrt{3}$, where $c, d \in \mathbb{Z}$ **2**
 - v. Find the smallest value $p > 0$ such that $(z_2)^p$ is a positive real number **1**
- b) Find $\int x^2 e^x dx$ **2**
- c) A mass has acceleration $a \text{ ms}^{-2}$ given by $a = v^2 - 3$, where $v \text{ ms}^{-1}$ is the velocity of the mass when it has a displacement of x metres from the origin. **4**
Find v in terms of x given that $v = -2$ where $x = 1$.

End of Question 12

Question 13 (15 marks) Begin a NEW page.

- a) The equations of intersecting lines L and M are given below with respect to a fixed origin O . **3**

$$L: \vec{r} = 11\vec{i} + 2\vec{j} + 17\vec{k} + \lambda(-2\vec{i} + \vec{j} - 4\vec{k})$$

$$M: \vec{r} = -5\vec{i} + 11\vec{j} + 1\vec{k} + \mu(p\vec{i} + 2\vec{j} + 2\vec{k})$$

where λ and μ are parameters and p is a constant.

If L and M are perpendicular, what is the value of p ?

- b) Prove by induction that **3**

$$\frac{1}{n!} < \frac{1}{2^{n-1}}$$

for $n \geq 3, n \in \mathbb{Z}^+$

- c) Let ω be a non-real cube root of unity

- i. Show that $1 + \omega + \omega^2 = 0$ **1**

- ii. Hence evaluate $(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2)$ **3**

- d) i. Show that **2**

$$\frac{\tan x}{7\sin^2 x + 5\cos^2 x} = \frac{\tan x \sec^2 x}{7\tan^2 x + 5}$$

- ii. Hence, by setting $s = \tan x$, or otherwise, find: **3**

$$\int_0^{\frac{\pi}{3}} \frac{\tan x dx}{7\sin^2 x + 5\cos^2 x}$$

End of Question 13

Question 14 (15 marks) Begin a NEW page.

- a) By writing $\frac{5x^2 - 3x + 1}{(x^2 + 1)(x - 2)}$ in the form $\frac{Ax + 1}{x^2 + 1} + \frac{B}{x - 2}$ find **3**

$$\int \frac{5x^2 - 3x + 1}{(x^2 + 1)(x - 2)} dx$$

- b) If a, b and $x \geq 0$,

i. Show that $a^2 + b^2 \geq 2ab$ **1**

ii. Hence, show that $\frac{x}{x^2 + 4} \leq \frac{1}{4}$ **2**

iii. By integrating both sides between the limits of $x = 0$ and $x = X$, show that **2**

$$e^{\frac{X}{2}} \geq \frac{X^2}{4} + 1 \text{ where } X \geq 0$$

- c) A particle is undergoing simple harmonic motion about $x = 0$. At time t seconds the displacement x in cm is given by

$$x = \sqrt{3} \sin 3t - \cos 3t$$

i. Write $x = \sqrt{3} \sin 3t - \cos 3t$ in the form of $x = A \sin(nt - \theta)$, where $A > 0$ **2**

ii. Find the period of the motion **1**

iii. When does the particle first reach maximum speed after time $t = 0$? **2**

iv. How long will it take for the particle to return to its original position, and find its acceleration at that point. **2**

End of Question 14

Question 15 (15 marks) Begin a NEW page.

- a) If xyz represents a three digit number (not the product of x, y and z), show that if $x + z = y$ then the number is divisible by 11. (x, y and z are positive integers) 2

- b) A particle is moving under simple harmonic motion where $v = 9\sqrt{3 - x^2}$. 2
Find the centre of motion.

- c) The points A and B have position vectors given by:

$$\vec{OA} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \text{ and } \vec{OB} = \begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix}$$

- i. Find an expression for the vector $\vec{AB}$ in the form of $x_1\tilde{i} + y_1\tilde{j} + z_1\tilde{k}$ 1
- ii. Show that the cosine of the angle between the vectors $\vec{OA}$ and $\vec{AB}$ is $\frac{4}{9}$ 2
- iii. Hence find the exact value of the area of ΔOAB 3

- d) i. Show that 1

$$\sin\left(\theta + \frac{\pi}{2}\right) = \cos\theta$$

- ii. Consider $f(x) = \sin ax$, where a is a constant. 4

Prove by mathematical induction that

$$f^{(n)}(x) = a^n \sin\left(ax + \frac{n\pi}{2}\right)$$

where $n \in \mathbb{Z}^+$ and $f^{(n)}(x)$ represents the n^{th} derivative of $f(x)$.

End of Question 15

Question 16 (15 marks) Begin a NEW page.

a) Let $I_n = \int_0^1 x^n e^{-x} dx$

i. Show that $I_n = nI_{n-1} - \frac{1}{e}$ **2**

ii. Hence or otherwise, find the exact value of $I_3 = \int_0^1 x^3 e^{-x} dx$ **2**

b) The point P representing the complex number z moves on the Argand diagram so that $|z| = |z - 6 + 4i|$

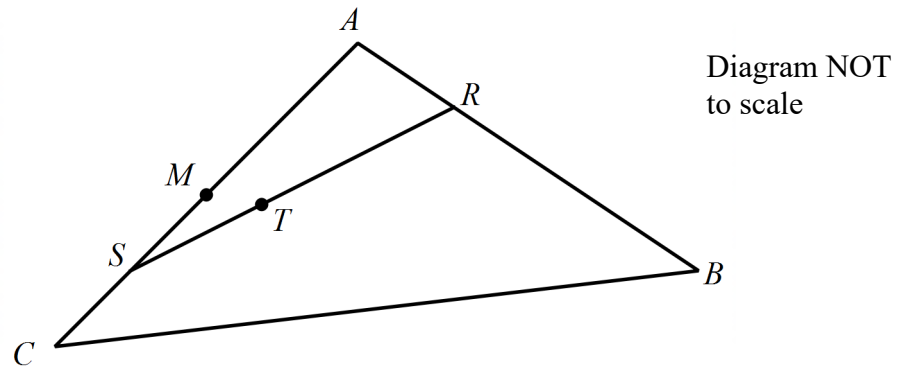
i. Find the Cartesian equation that describes the locus of P **2**

ii. Graph the locus neatly on a number plane. **1**

iii. Hence find the minimum value of $|z|$ **1**

Question 16 continues on the next page

- c) The position vectors of the points A, B and C are $\vec{a}, \vec{b}$ and $\vec{c}$ respectively, relative to an origin O . The following diagram shows the triangle ABC and the points M, R, S and T .



M is the midpoint of $\vec{AC}$
 R is on $\vec{AB}$ such that $\vec{AR} = \frac{1}{3} \vec{AB}$
 S is on $\vec{AC}$ such that $\vec{AS} = \frac{2}{3} \vec{AC}$
 T is a point on $\vec{RS}$ such that $\vec{RT} = \frac{2}{3} \vec{RS}$

- | | | |
|------|---|----------|
| i. | Express $\vec{AM}$ in terms of $\vec{a}$ and $\vec{c}$ | 1 |
| ii. | Hence show $\vec{BM} = \frac{1}{2} \vec{a} - \vec{b} + \frac{1}{2} \vec{c}$ | 1 |
| iii. | Show that $\vec{RT} = -\frac{2}{9} \vec{a} - \frac{2}{9} \vec{b} + \frac{4}{9} \vec{c}$ | 2 |
| iv. | Prove that T lies on $\vec{BM}$ | 3 |

End of Exam

Extension 2 Mathematics Trial Solutions

Section 1.

1. B 6. A

2. A 7. B

3. C 8. C

4. A 9. A

5. A 10. A

$$1. (2e^{i\pi/3})^3 = 8e^{i\pi}$$

Purely real.

$$2. n = \sqrt{3} \\ T = \frac{2\pi}{\sqrt{3}}$$

Max $\rightarrow$ min is $\frac{1}{2}T$

$$\therefore \frac{\pi}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\pi}{3} \text{ seconds}$$

$$3. a = -c \\ a + c = 0$$

$$4. \frac{1}{1-(3-4i)} = \frac{1}{-2+4i}$$

$$\frac{1}{-2+4i} \times \frac{-2-4i}{-2-4i}$$

$$= \frac{-2(1+2i)}{4+16}$$

$$= \frac{1+2i}{10}$$

$$5. a > b$$

$$a - b > 0$$

$$\frac{1}{a-b} > 0$$

$$6. a \cdot b = 0$$

$$m^2 + 4m - 12 = 0$$

$$(m-2)(m+6) = 0$$

$$7. x = 2m+1, y = 2n+1$$

$$x - y = 2m+1 - (2n+1) = 2m - 2n$$

$$= 2(m-n)$$

$$8. \text{ let } u = \ln x \Rightarrow du = \frac{1}{x} dx$$

$$x = e^3 \Rightarrow u = 3$$

$$x = 1 \Rightarrow u = 0$$

$$\therefore I = \int_0^3 u^3 du$$

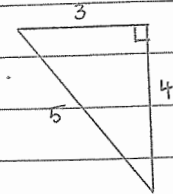
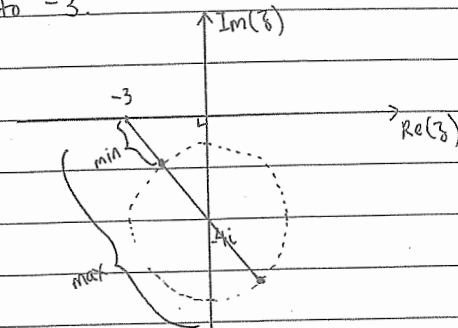
$$9. |z - (-4i)| = 3$$

is a circle.

where $|z+3| = |z-(-3)|$
magnitude of a vector

joining point on the circle

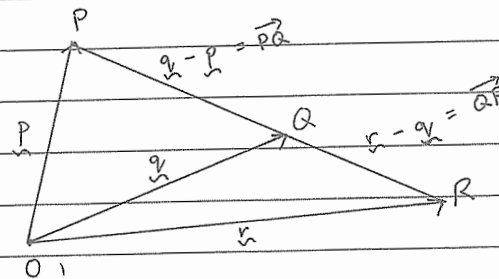
to -3.



min value : $5 - 3$

max value : $5 + 3$
radius

10.



$$2(r - q) = q - p$$

$$2r = 3q - p$$

$$r = \frac{3}{2}q - \frac{1}{2}p$$

Section II.

11. a) $|z| = \sqrt{6^2 + 3^2 + 1^2}$

$$\hat{z} = \frac{1}{\sqrt{46}} \begin{pmatrix} 6 \\ 3 \\ -1 \end{pmatrix}$$

b) i. $z = 6 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$

ii. $z^4 = \left[6 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \right]^4$

By De Moivre's Theorem.

$$z^4 = 6^4 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right)$$

$$= 6^4 \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right)$$

OR $6^4 \left(\cos \left(-\frac{2\pi}{3} \right) \right)$

iii. $\arg(iz) = \arg(i) + \arg(z)$
 $= \frac{\pi}{2} + \frac{\pi}{3}$
 $= \frac{5\pi}{6}$

c) i. $\int \frac{x+1}{x^2+2x+5} dx = \frac{1}{2} \int \frac{2x+2}{x^2+2x+5} dx$
 $= \frac{1}{2} \ln |x^2+2x+5| + C$

ii. $\int \frac{1}{x^2+2x+5} dx = \int \frac{1}{(x^2+2x+1)+4} dx$
 $= \int \frac{1}{(x+1)^2+4} dx$
 $= \frac{1}{2} \tan^{-1} \left(\frac{x+1}{2} \right) + C$

d) i. If it is not a doggo, then/it implies that it is (not) a woofier.

ii. $(\text{not } q \text{ and } r) \Rightarrow p$

iii. $\text{not}(\text{not } q) \Rightarrow \text{not } p$
 $q \Rightarrow \text{not } p$

12. a) i. $z_1 = 1 + i$

$$|z_1| = \sqrt{2}$$

$$\arg(z_1) = \frac{\pi}{4}$$

$$\therefore z_1 = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

ii. $|z_1 z_2| = |z_1| |z_2|$

$$= 2 \times \sqrt{2}$$

$$= 2\sqrt{2}$$

iii. $z_1 z_2 = 2\sqrt{2} \left(\cos \left(\frac{\pi}{6} + \frac{\pi}{4} \right) + i \sin \left(\frac{\pi}{6} + \frac{\pi}{4} \right) \right)$

$$\cos \left(\frac{\pi}{6} + \frac{\pi}{4} \right) = \cos \frac{\pi}{6} \cos \frac{\pi}{4} - \sin \frac{\pi}{6} \sin \frac{\pi}{4}$$

$$= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}$$

$$\sin \left(\frac{\pi}{6} + \frac{\pi}{4} \right) = \sin \frac{\pi}{6} \cos \frac{\pi}{4} + \cos \frac{\pi}{6} \sin \frac{\pi}{4}$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}}$$

$$\therefore z_1 z_2 = 2\sqrt{2} \left(\frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} + i \left(\frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}} \right) \right)$$

$$= \sqrt{3} - 1 + i(1 + \sqrt{3})$$

iv. $\tan \frac{5\pi}{12} = \frac{\sin \frac{5\pi}{12}}{\cos \frac{5\pi}{12}}$

$$= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{3 + 2\sqrt{3} + 1}{3 - 1}$$

$$= 2 + \sqrt{3}$$

v. $(z_2)^p$ positive real. $\cos \frac{p\pi}{6} > 0$

$$\sin \frac{p\pi}{6} = 0$$

$$\therefore p = 12$$

$$b) \int x^2 e^x dx$$

$$= [x^2 e^x] - \int 2x e^x dx$$

$$= x^2 e^x - ([2x e^x] - \int 2e^x dx)$$

$$= x^2 e^x - 2x e^x + 2e^x + C$$

$$c) \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$v^2 - 3 = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \quad \text{OR}$$

$$2v^2 - 6 = \frac{dv^2}{dx}$$

$$\frac{1}{2v^2 - 6} = \frac{dx}{dv^2}$$

$$v^2 - 3 = v \frac{dv}{dx}$$

$$\frac{v^2 - 3}{v} = \frac{dv}{dx}$$

$$\int \frac{dv^2}{2v^2 - 6} = \int dx$$

$$\int \frac{v}{v^2 - 3} dv = \int dx$$

$$x = \frac{1}{2} \ln |2v^2 - 6| + C \quad \frac{1}{2} \ln |v^2 - 3| = x + C$$

$$\text{When } x = 1, v = -2$$

$$\text{When } x = 1, v = -2$$

$$1 = \frac{1}{2} \ln |8 - 6| + C$$

$$C = -1$$

$$1 = \frac{1}{2} \ln 2 + C$$

$$\Rightarrow \frac{1}{2} \ln |v^2 - 3| = x - 1$$

$$C = 1 - \frac{1}{2} \ln 2$$

$$x = \frac{1}{2} \ln (2v^2 - 6) + 1 - \frac{1}{2} \ln 2$$

$$= \frac{1}{2} \ln |v^2 - 3| + 1$$

$$2(x-1) = \ln |v^2 - 3|$$

$$e^{2(x-1)} = v^2 - 3$$

$$v^2 = e^{2(x-1)} + 3$$

$$v = \pm \sqrt{e^{2(x-1)} + 3}$$

$$\text{but as } x = 1, v = -2$$

$$\Rightarrow v = -\sqrt{e^{2(x-1)} + 3} \text{ only.}$$

13. a) If perpendicular, direction vectors must be perpendicular
i.e. dot product zero.

$$(-2\hat{i} + \hat{j} - 4\hat{k}) \cdot (p\hat{i} + 2\hat{j} + 2\hat{k}) = 0$$

$$-2p + 2 - 8 = 0$$

$$-2p = 6$$

$$p = -3$$

$$b) \frac{1}{n!} < \frac{1}{2^{n-1}}, \quad n \geq 3, \quad n \in \mathbb{Z}^+$$

Show true for $n=3$.

$$\text{LHS} = \frac{1}{3!}$$

$$= \frac{1}{6}$$

$$\text{RHS} = \frac{1}{2^{3-1}}$$

$$= \frac{1}{4} > \frac{1}{6}$$

$\therefore$ True for $n=3$.

OR:

Assume true for some $k \in \mathbb{N}$

$$\text{LHS} = \frac{1}{(k+1)!}$$

$$\text{i.e. } \frac{1}{k!} < \frac{1}{2^{k-1}}$$

$$= \frac{1}{k! (k+1)}$$

Now prove true for $n=k+1$

$$< \frac{1}{2^{k-1}} \cdot \frac{1}{(k+1)} \text{ assump.}$$

$$\text{i.e. Prove } \frac{1}{(k+1)!} < \frac{1}{2^{k+1-1}}$$

$$= \frac{1}{k 2^{k-1} + 2^k}, \quad k \geq 3$$

From the assumption:

$$< \frac{1}{2 \cdot 2^{k-1} + 2^k}$$

$$\frac{1}{k!} \times \frac{1}{k+1} < \frac{1}{2^{k-1}} \times \frac{1}{k+1}$$

$$= \frac{1}{3 \cdot 2^{k-1}}$$

$$\frac{1}{(k+1)!} < \frac{1}{2^{k-1} (k+1) \times 2}$$

$$< \frac{1}{2 \cdot 2^{k-1}}$$

$$< \frac{1}{2^{k-1} (k+1) \times 2}$$

$$= \frac{1}{2^{k-1+1}}$$

$$< \frac{1}{2^{k+1-1} (k+1)}$$

$$= \frac{1}{2^{k+1-1}}$$

$$< \frac{1}{2^{k+1-1}} \times \frac{2}{k+1}$$

$$\therefore \frac{1}{(k+1)!} < \frac{1}{2^{k+1-1}}$$

$$\frac{1}{(k+1)!} < \frac{1}{2^{k+1-1}} \quad 0 < \frac{2}{k+1} < 1 \text{ for } k \geq 3$$

$\therefore$ True for $n=k+1$, given true for $n=k$

Since result is true for $n=3$, it follows that it will

also be true for $n=3+1=4$, $n=4+1=5$ and so on for all $n \geq 3$ integers.

c) i. $z^3 = 1$

$$z^3 - 1 = 0$$

$$(z-1)(z^2+z+1)=0$$

Now ω is a non-real root of unity.

i.e. $(\omega-1)(\omega^2+\omega+1)=0$ but $\omega \neq 1$

$$\therefore \omega^2 + \omega + 1 = 0.$$

ii. $(1-3\omega+\omega^2)(1+\omega-8\omega^2)$

$$= ((1+\omega^2)-3\omega)((1+\omega)-8\omega^2)$$

$$= (-\omega-3\omega)(-\omega^2-8\omega^2)$$

$$= -4\omega \times -9\omega^2$$

$$= 36\omega^3 \text{ but } \omega^3 = 1$$

$$= 36$$

d) i.
$$\text{LHS} = \frac{\tan x}{7\sin^2 x + 5\cos^2 x}$$

$$= \frac{\frac{\tan x}{\cos^2 x}}{7 \frac{\sin^2 x}{\cos^2 x} + 5 \frac{\cos^2 x}{\cos^2 x}}$$

$$= \frac{\tan x \sec^2 x}{7\tan^2 x + 5}$$

$$= \text{RHS}$$

ii.
$$\int_0^{\pi/3} \frac{\tan x}{7\sin^2 x + 5\cos^2 x} dx$$

$$= \int_0^{\pi/3} \frac{\tan x \sec^2 x}{7\tan^2 x + 5} dx$$

from part i)

Let $s = \tan x$

$$ds = \sec^2 x dx$$

when $x = \pi/3$, $s = \sqrt{3}$

$x = 0$, $s = 0$

$$\therefore I = \int_0^{\sqrt{3}} \frac{s ds}{7s^2 + 5}$$

$$= \frac{1}{14} \left[\ln |7s^2 + 5| \right]_0^{\sqrt{3}}$$

$$= \frac{1}{14} (\ln 26 - \ln 5)$$

$$= \frac{1}{14} \ln \frac{26}{5}$$

$$14 \text{ a) } \frac{5x^2 - 3x + 1}{(x^2 + 1)(x - 2)} = \frac{Ax + 1}{x^2 + 1} + \frac{B}{x - 2}$$

$$5x^2 - 3x + 1 = (Ax + 1)(x - 2) + B(x^2 + 1)$$

$$\text{let } x = 2$$

$$15 = B(5)$$

$$B = 3$$

$$\text{let } x = 1$$

$$3 = (A + 1)(-1) + 3(2)$$

$$-3 = -A - 1$$

$$-2 = -A$$

$$A = 2$$

$$I = \int \frac{2x + 1}{x^2 + 1} + \frac{3}{x - 2} dx$$

$$= \int \frac{2x}{x^2 + 1} + \frac{1}{x^2 + 1} + \frac{3}{x - 2} dx$$

$$= \ln|x^2 + 1| + \tan^{-1}x + 3\ln|x - 2| + C$$

$$= \ln|(x^2 + 1)(x - 2)^3| + \tan^{-1}x + C$$

$$\text{b) i. } (a - b)^2 \geq 0$$

$$a^2 - 2ab + b^2 \geq 0$$

$$a^2 + b^2 \geq 2ab$$

$$\text{ii. let } a = x, b = 2$$

$$x^2 + 4 \geq 4x$$

$$\frac{1}{x^2 + 4} \leq \frac{1}{4x}, \quad x \text{ is positive so}$$

$$\frac{x}{x^2 + 4} \leq \frac{1}{4}$$

$$\text{iii. } \int_0^x \frac{x}{x^2 + 4} dx \leq \int_0^x \frac{1}{4} dx$$

$$\left[\frac{1}{2} \ln|x^2 + 4| \right]_0^x \leq \left[\frac{1}{4}x \right]_0^x$$

$$\frac{1}{2} \ln|x^2 + 4| - \frac{1}{2} \ln 4 \leq \frac{1}{4}x$$

$$\frac{1}{2} \ln \frac{x^2 + 4}{4} \leq \frac{x}{4}$$

$$\ln \frac{x^2 + 4}{4} \leq \frac{x}{2}$$

$$e^{\ln \left(\frac{x^2 + 4}{4} \right)} \leq e^{\frac{x}{2}}$$

$$\frac{x^2}{4} + 1 \leq e^{\frac{x}{2}}$$

$$\text{c) i. } x = \sqrt{3} \sin 3t - \cos 3t$$

$$x = A \sin 3t \cos \theta - A \cos 3t \sin \theta$$

$$\Rightarrow A \cos \theta = \sqrt{3}$$

$$A \sin \theta = 1$$

$$A = 2$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6}$$

$$\therefore x = 2 \sin(3t - \frac{\pi}{6})$$

$$\text{ii. } T = \frac{2\pi}{n}, \quad n = 3$$

$$T = \frac{2\pi}{3}$$

$$\text{iii. Max speed at } x = 0.$$

$$2 \sin(3t - \frac{\pi}{6}) = 0$$

$$\text{i.e. } 3t - \frac{\pi}{6} = 0, \pi, 2\pi, \dots$$

$$\text{Take } 3t - \frac{\pi}{6} = 0$$

$$3t = \frac{\pi}{6}$$

$$t = \frac{\pi}{18} \text{ seconds.}$$

iv. Original position : $t=0$.

$$x = 2 \sin\left(-\frac{\pi}{6}\right)$$

$$= 2 \times -\frac{1}{2}$$

$$= -1$$

$$\therefore -1 = 2 \sin\left(3t - \frac{\pi}{6}\right)$$

$$-\frac{1}{2} = \sin\left(3t - \frac{\pi}{6}\right)$$

$$\Rightarrow 3t - \frac{\pi}{6} = -\frac{\pi}{6} \text{ or } \frac{7\pi}{6} \text{ or } \dots$$

↑
this is $t=0$.

$$\therefore 3t - \frac{\pi}{6} = \frac{7\pi}{6}$$

$$3t = \frac{8\pi}{6}$$

$$t = \frac{4\pi}{8 \times 3}$$

$$= \frac{4\pi}{9} \text{ seconds}$$

Acceleration at the point $t = \frac{4\pi}{9}$, $x = -1$

$$\ddot{x} = -\omega^2 x$$

$$= -3^2 x - 1$$

$$= 9 \text{ cm/s}^2$$

15. a) $xyz = 100x + 10y + z$

$$\text{but } y = x + z$$

$$\therefore xyz = 100x + 10(x+z) + z$$

$$= 110x + 11z$$

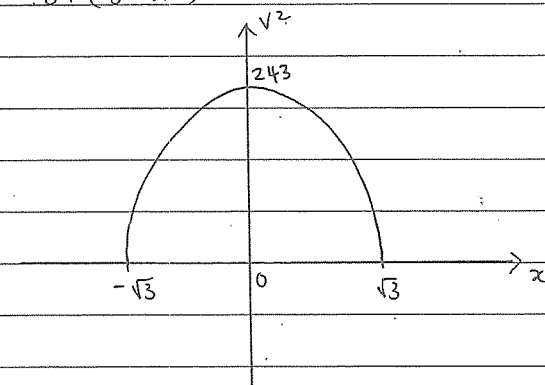
$$= 11(10x + z)$$

$\therefore$ divisible by 11 as x, z are positive integers.

b) $v = 9\sqrt{3-x^2}$

(centre is where v is max, v^2 is max.)

$$v^2 = 81(3-x^2)$$



Centre of motion at $x=0$.

c) i. $\vec{AB} = -1\hat{i} + 3\hat{j} + 4\hat{k} - (1\hat{i} + 2\hat{j} + 2\hat{k})$

$$= -2\hat{i} + \hat{j} + 2\hat{k}$$

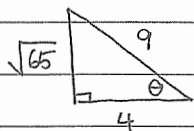
ii.
$$\cos \theta = \frac{\vec{OA} \cdot \vec{AB}}{|\vec{OA}| \times |\vec{AB}|}$$

$$= \frac{-2 + 2 + 4}{\sqrt{9} \times \sqrt{9}}$$

$$= \frac{4}{9}$$

$$\text{iii. } \Delta OAB = \frac{1}{2} |\vec{OA}| \times |\vec{OB}| \sin \theta$$

$$= \frac{1}{2} \times 9 \times \sin \theta$$



$$\Delta OAB = \frac{1}{2} \times 9 \times \frac{\sqrt{65}}{9}$$

$$= \frac{1}{2} \sqrt{65} u^2$$

$$\text{d) i. } \sin\left(\theta + \frac{\pi}{2}\right) = \sin \theta \cos \frac{\pi}{2} + \cos \theta \sin \frac{\pi}{2}$$

$$= \sin \theta \times 0 + \cos \theta \times 1$$

$$= \cos \theta$$

$$\text{ii. } f^{(n)}(x) = a^n \sin\left(ax + \frac{n\pi}{2}\right)$$

Show true for $n=1$

$$f'(x) = a \cos ax$$

$$\text{By formula: } f'(x) = a' \sin\left(ax + \frac{\pi}{2}\right)$$

$$= a \cos ax \quad (\text{by part i.})$$

Assume true for $n=k$

$$\text{i.e. } f^k(x) = a^k \sin\left(ax + \frac{k\pi}{2}\right)$$

Prove true for $n=k+1$

$$\text{i.e. } f^{(k+1)}(x) = a^{(k+1)} \sin\left(ax + \frac{(k+1)\pi}{2}\right)$$

From the assumption:

$$f^{(k)}(x) = a^k \sin\left(ax + \frac{k\pi}{2}\right)$$

differentiate both sides

$$f^{(k+1)}(x) = a^k \cos\left(ax + \frac{k\pi}{2}\right) \times a$$

$$= a^{k+1} \cos\left(ax + \frac{k\pi}{2}\right)$$

$$= a^{k+1} \sin\left(ax + \frac{k\pi}{2} + \frac{\pi}{2}\right) \quad (\text{by part i.})$$

$$= a^{k+1} \sin\left(ax + (k+1)\frac{\pi}{2}\right)$$

$\therefore$ true for $n=k+1$ provided $n=k$ is true.

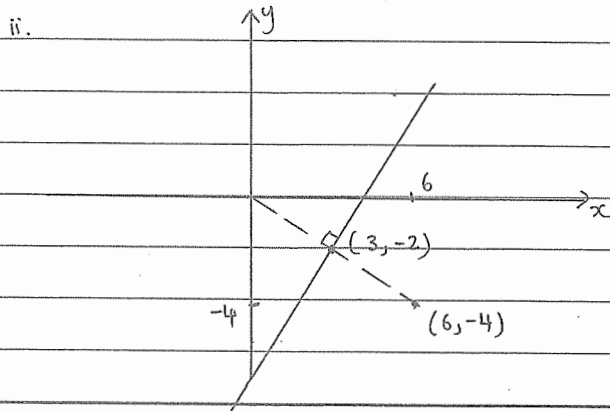
Since true for $n=1$, then it follows true for $n=1+1=2$,
 $n=2+1=3$ and so on for all integers $n \geq 1$.

16. a) $I_n = \int_0^1 x^n e^{-x} dx$

i. $I_n = [x^n \times -e^{-x}]_0^1 - \int_0^1 nx^{n-1}(-e^{-x}) dx$
 $= 1^n \times (-e^{-1}) - 0^n \times (-e^0) + \int_0^1 nx^{n-1} e^{-x} dx$
 $= -\frac{1}{e} + n \int_0^1 x^{n-1} e^{-x} dx$
 $= -\frac{1}{e} + n I_{n-1}$

ii. $I_3 = 3I_2 - \frac{1}{e}$
 $= 3(2I_1 - \frac{1}{e}) - \frac{1}{e}$
 $= 6I_1 - \frac{4}{e}$
 $= 6(I_0 - \frac{1}{e}) - \frac{4}{e}$
 $= 6I_0 - \frac{10}{e}$
 $= 6 \int_0^1 x^0 e^{-x} dx - \frac{10}{e}$
 $= 6 \int_0^1 e^{-x} dx - \frac{10}{e}$
 $= 6[-e^{-x}]_0^1 - \frac{10}{e}$
 $= 6(-e^{-1} + 1) - \frac{10}{e}$
 $= -\frac{6}{e} + 6 - \frac{10}{e}$
 $= 6 - \frac{16}{e}$

b) i. $|z| = |8 - 6 + 4i|$
 $= |8 - (6 - 4i)|$ perpendicular bisector



i. $M = (3, -2)$

$m = -\frac{4}{6} \Rightarrow$ perpendicular gives $\frac{3}{2}$

$\therefore y + 2 = \frac{3}{2}(x - 3)$

$y = \frac{3}{2}x - \frac{9}{2} - 2$
 $= \frac{3}{2}x - \frac{13}{2}$

OR $3x - 2y - 13 = 0$

iii. Minimum value by Pythagoras' Theorem:

$\sqrt{3^2 + 2^2} = \sqrt{13}$

c) i. $\vec{AM} = \frac{1}{2}(\vec{c} - \vec{a})$

ii. $\vec{BM} = \frac{1}{2}\vec{a} - \vec{b} + \frac{1}{2}\vec{c}$

$= \vec{OM} - \vec{OB}$

$= \vec{OA} + \vec{AM} - \vec{OB}$

$= \vec{a} + \frac{1}{2}(\vec{c} - \vec{a}) - \vec{b}$

$= \frac{1}{2}\vec{a} - \vec{b} + \frac{1}{2}\vec{c}$

iii. $\vec{RT} = \vec{OT} - \vec{OR}$

$\vec{RT} = \frac{2}{3}\vec{RS}$

$= \frac{2}{3}(\vec{OS} - \vec{OR})$

$= \frac{2}{3}([\vec{OC} + \frac{1}{3}\vec{CA}] - [\vec{OB} + \frac{2}{3}\vec{BA}])$

$= \frac{2}{3}(\vec{c} + \frac{1}{3}(\vec{a} - \vec{c}) - \vec{b} - \frac{2}{3}(\vec{a} - \vec{b}))$

$= \frac{2}{3}(\frac{2}{3}\vec{c} - \frac{1}{3}\vec{a} - \frac{1}{3}\vec{b})$

$= \frac{4}{9}\vec{c} - \frac{2}{9}\vec{a} - \frac{2}{9}\vec{b}$

iv. Collinear:

Prove $\vec{BT} = \lambda \vec{BM}$

$\vec{BT} = \lambda(\frac{1}{2}\vec{a} - \vec{b} + \frac{1}{2}\vec{c})$ (from part ii)

Now $\vec{BT} = \vec{BR} + \vec{RT}$

$= -\frac{2}{3}\vec{AB} + \vec{RT}$

$$= -\frac{2}{3} \left(\frac{1}{2}a - a \right) + \left(\frac{4}{9}a - \frac{2}{9}a - \frac{2}{9}b \right) \text{ (from part iii)}$$

$$= -\frac{2}{9}b + \frac{4}{9}a + \frac{4}{9}a$$

$$= \frac{4}{9} (a - 2b + a)$$

$$= \frac{8}{9} \left(\frac{1}{2}a - b + \frac{1}{2}a \right)$$

$$= \frac{8}{9} \vec{BM}$$

$\therefore T$ lies on $\vec{BM}$.